

Background

Little Red Dots are compact, red sources seen by JWST. Their spectra show broad Balmer lines and blackbody-like continua, puzzling astronomers.

Their inferred black hole masses may be wrong, as the emission could arise from dense winds, not virial motion.

What this paper is selling

This paper draws parallels with Type IIn supernovae and Eta Carinae's Great Eruption. It proposes that LRDs are super-Eddington, intermediate-mass engines enshrouded in dense gas, explaining their observed features.

Key results to remember

- LRDs cluster at ~5000 K, like IIn SNe and Eta Carinae.
- Wind physics mimics virial broad lines, overestimating black hole masses.
- Engine mass constrained to <1e5 solar masses, super-Eddington.

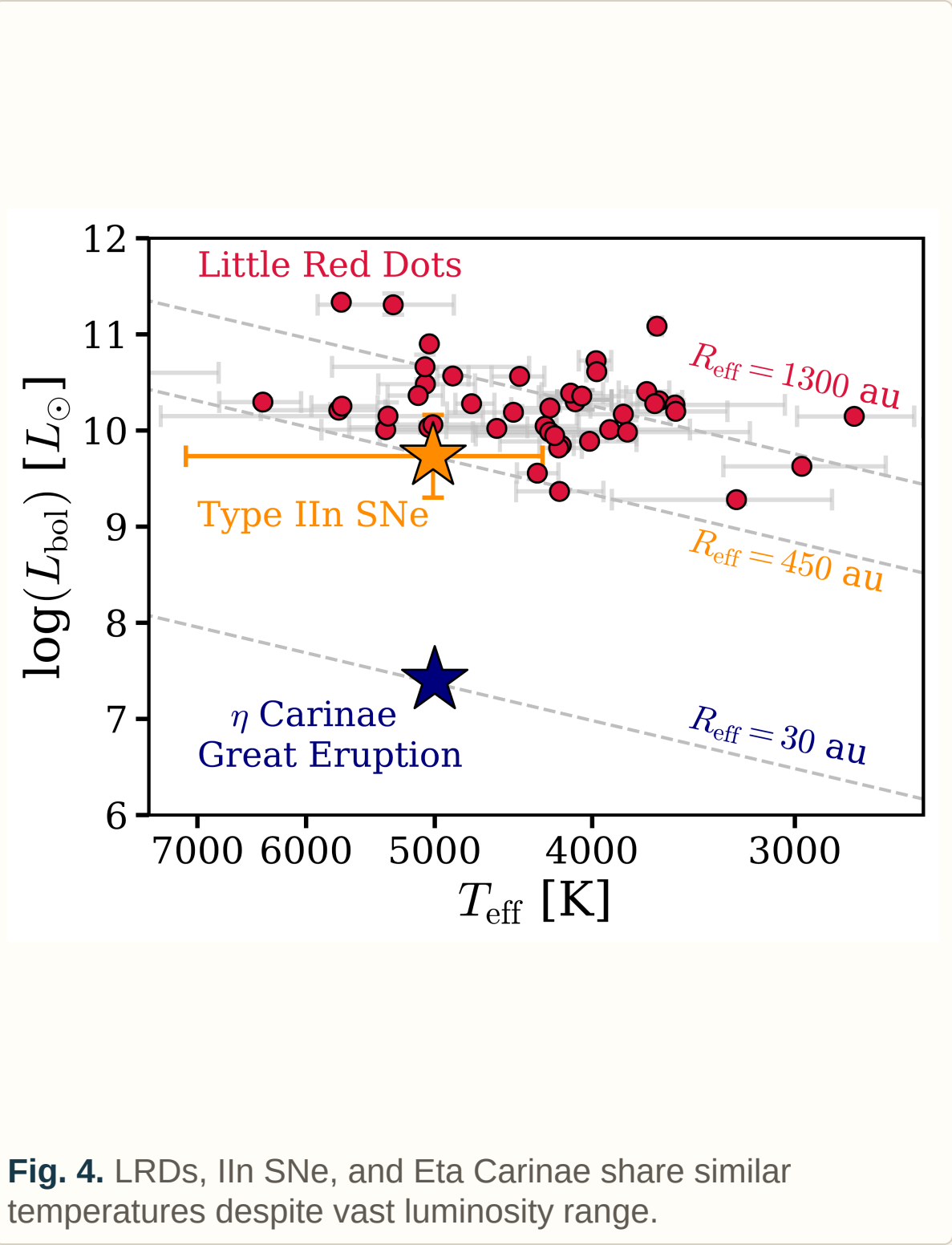


Fig. 4. LRDs, IIn SNe, and Eta Carinae share similar temperatures despite vast luminosity range.

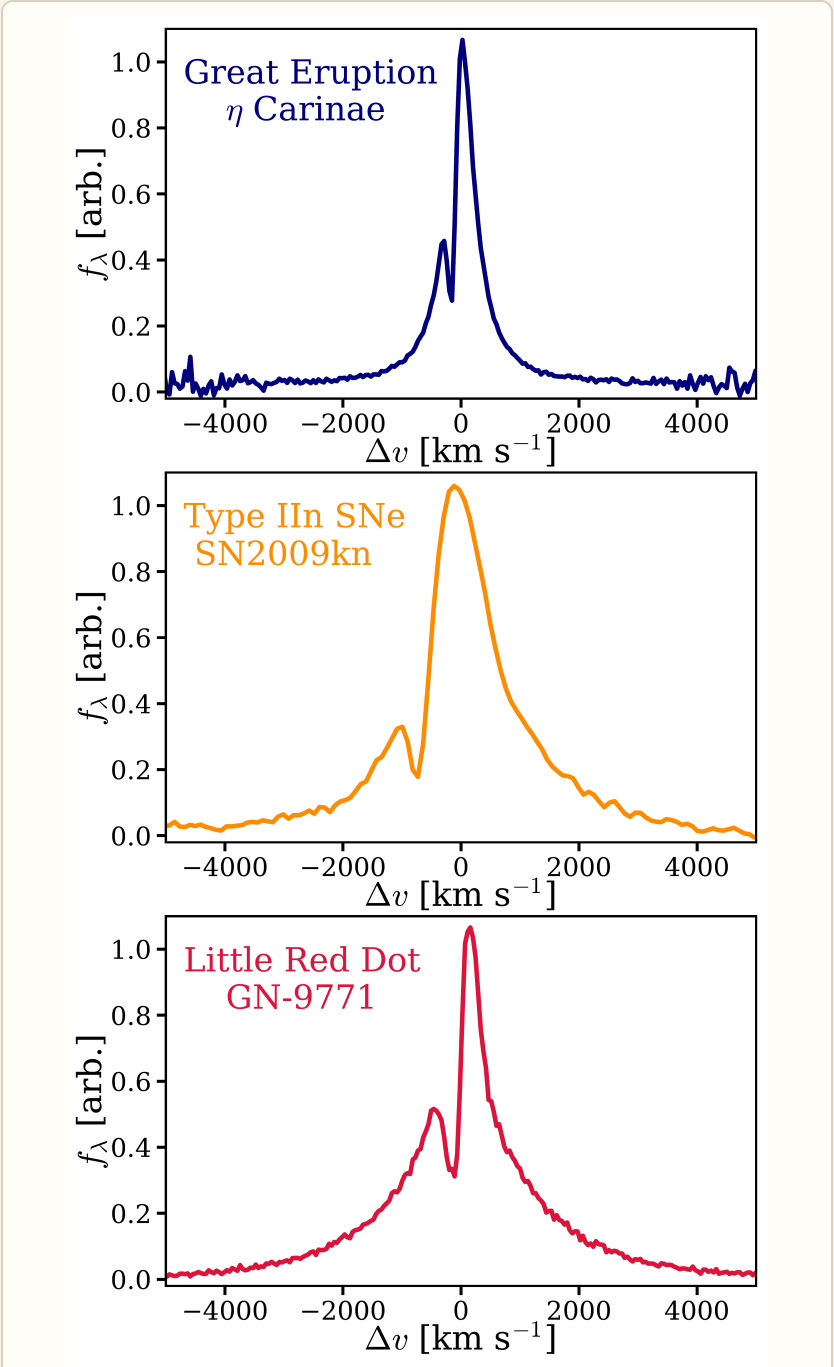


Fig. 1. Striking similarity in H-alpha profiles: electron-scattered wings and P-Cygni absorption.